

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 3

Source: [YouTube](#) **Playlist:** Physics as Information Processing ~ Chris Fields ~ 2023 **Duration:** 1:10:29 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome it's July 29th 2023 and we're here in session three discussion of the course physics this information processing at the Active inference Institute we're having a follow-up discussion on Chris fields's third lecture on Quantum reference frames and we're looking forward to your participation so if you're watching along live and you've been invited to the calendar event it's not too late to jump in if you're watching on YouTube you can write any questions or comments in the live chat with Dean and Andrew here we're going to be looking at some of the key points and taking it in a few different directions sharing our reference frames making a shared one so enter be awesome to have you zoom in a little bit on the slide and then feel free to take us through some of the key points so that we can um see a little bit how you're seeing it thank you that looks good yep okay so let's let's start by reviewing um the last lecture from a couple weeks ago and then we can open up the discussions if there's any questions that appear in the live chat we can address those uh all right so lecture two Chris introduced this idea of uh the information processing between system and you know where they interact via qubits on a boundary and they have um reference frames that give meaning to this com to this communication you want to think of uh information processing as a communication Channel uh the way um Shannon thought about it uh so that was the lecture too and lecture three a couple weeks ago uh try to make this more explicit by emphasizing the idea that this has to be physically embodied somewhere okay and I think that because we are so used to information being fungible in no small part because of the internet and everything uh it's it's a little hard uh to understand at first when the claims made that you have a qubit that it's non-fungible or you have a piece of quantum information that's not fungible or things like the no cloning theorem we're so used information being punishable to it being zeros and ones that it's a bit of a change in in mindset at first but that's exactly what lecture three um introduced this was all all stuff right um could you maybe just unpack that a little bit what would it mean for information to be fungible or what's that stance like and then what does it

mean for information to be non-fungible and what is that stance like yeah I'm about to get there uh so let me let me scroll down the slides um because that's addressed here so I was maybe I should have kept this uh in the beginning but anyways session two talked about how information Theory makes this simple and obvious okay yeah session three we talk about reference frames and mechanistically how all of this works um so yeah let's address that question of the difference between hold on one second let me use my mouse to scroll down here okay so that's that's a little faster right okay so can you restate your question uh Daniel yeah what would it mean for information to be fungible or non-fungible people might be familiar with that in the economic setting with non-fungible tokens and all of that but what are we talking about with informational fungibility I think it basically comes down to um to whether you can replicate I mean it's basically the difference between classical and Quantum information as far as I know uh uh uh fungible information is everything that you can basically turn into a message of zeros and ones uh it's you know how you transmit it maybe different you know the channel the physical hardware and so on but but at the end of the day it's something that you can write on a boundary and something that you can transmit at zero seven ones whereas Quantum information [Music] um you literally have to send the hardware the piece of Hardware uh the qrf at least that's how I understand it so a simple answer might be um that it's information encoded in a piece of software versus a piece of Hardware but I don't I don't think that analogy is perfect but what do you yeah Dean what do you think well here's what I'm curious about and again you guys can help me with this as opposed to reference frames I oftentimes default to frame of reference and I know then one of the live streams what I tried to bring out was the idea that there was an ice cream cone looking in an ice cream truck and the question was always whether or not which which one of those two entities is one is in the foreground and one is in the background and in this slide that Ander has a the question is is the non-fungibility because only A and B can decide whether the a the VA as seen by B the a is deeper into that image be based on our frame of reference and the B is closer or is the B the one that's actually deeper and the a is the one that's in the foreground and I was wondering whether the idea of non-copyability is because of that depth of field aspect that cannot be determined unless it's literally handed over the way that Chris was describing it I think I think that's basically accurate I think um I'm not sure I can answer the question but I think I have the feeling that Chris well Chris said this in the lecture that I mean is somewhere in your your nervous system and uh here whether I mean you bring up a good point right it looks like this slide is a bit of a an optical illusion whether you see A or B deeper uh I guess it depends on

something going on inside your head um if you this picture is so symmetrical that if you focus on albumin you can change your opinion um what would just to even connect that to what we're talking about the classical fungible information on the holographic screen on the blanket and then the quantum cognitive rotation is those two images are on the screen identical and then whether it gets a Quantum rotation such that the uh three points of the axes are closer to us than the intersection on that kind of tetrahedron or the other way around whether the ballerina is rotating one way or the other that's not on the boundary so the information about the depth of field is part of the qrf or part of the enactment of the qrf and it's not written on the boundary and that goes back to the preparation question and not just the measuring yeah like if the preparation if your whole enculturation or the last five minutes it was uh entrained or you were holding an object that predisposed you to think that the vertex of the intersection was further away that's the preparation that's your that's your prior so that when you get this ambiguous artifact you rotate it into the most natural position cognitively and that's the preparation that sets you up for measurement no measurement can happen without a preparation what do you think Andrew uh sure yeah everything is that makes sense so uh perhaps let me let me recall what was that around the slides uh there was a in this whole Spirit or philosophy of making emphasizing on information is physical you know inconcubing this whole thing for me um one of the aglarian emissions in in physics was the embodiment of the reference frame which was treated as an abstraction um you know in classical physics such as by galindale but maybe even Einstein even if he my hunch is that he did have more of this Instinct of you know particularly in all those thought experiments of trying to chase the library you are chasing the library right so at um but I mean this this is a whole process of making the Observer and the reference frame more and more explicit and more and more physical because at the end of the day it has to right if if you start with this assumption that solipsistic assumption almost that you know all you see and and you know is is part of the same material uh ether that you are then you know then you are part of the universe that in Europe serving something that is out there uh so you need to make include a theory of what it means for you to be observing that right and at the end of the day the Observer has to be physical that's what was emphasized by by Chris at the beginning of this lecture so this jump from treating the reference frame as an abstraction to a real physical entity and yeah these are these people have uh about uh in these papers that are ordered there but I should also remark uh if folks want to see this it's it's coming up in high energy Theory also in the recent work of um in recent work of saskins and others where they talk about Quantum reference traits uh very explicit so I don't

I don't have the citations off the bat but it's definitely becoming an important thing in the last 20 years this attempt to really explicitly treat the reference frame on the Observer asset Quantum entity uh as a physical entity okay so this that's what I just said okay the jump from treating things reference rooms as abstractions to physical objects go and because we cannot make a perfect copy of quantum information we say that these are unique and clonable and unfungible um so in order to recreate an experiment perfectly uh this is the this is the address as a part of fungibility or not right Canalis if you have some sort of Quantum experiment and Analysis involved are you know it's the usual DPR uh thought experiment or whatever Alice involved are separated by light years is it enough if you want to recreate the same exact the same experiment for Alice to send string of zeros and ones well if you had an experiment where that's enough then then you can recreate it with tangible information if not then you need to somehow send non-tangible information which really implies in a piece of hardware um can I can I add another uh another piece here look on the previous uh slide so this looks really Technical and scientific but let's think about Alice's laboratory is Alice's brain mind and Bob's analogously Alice is thinking about what to say and then says it and that information might be written in a chat form it might be said verbally it might be transmitted through digital or analog signal but at some point it's going through this classical informational interface and then Bob's laboratory is like perceiving and unpacking it and the question of semantic alignment is such that if Alice gives a shopping list of 10 items if there's semantic alignment Bob would be able to go to the store and then find those 10 items and it made me think of the the dictionary definition of left and right so this gets back to our discussion of whether the vertex on these axes is coming towards us or away from us so I looked up the dictionary definition of left on towards or relating to the side of a human body or of a thing that is to the West when the person or that thing is facing north so that's I guess one attempt to Anchor what it means to be left the second definition is the left hand part side or direction and you could imagine there's an analogous definition for right the point is you can't describe verbally leftness and rightness or chirality in general because its reference is an embodied perspective and so if Alice said okay you're gonna go down the street and then turn left at the stop sign how would Alice communicate to Bob what leftness meant at that stop sign versus rightness and that's this question of flipping and the way that the quantum formalism accommodates that is through the wick rotation and the quantum cognitive events where the fungible transmittable holographic information gets rotated into the imaginary axis with the formalisms that enable it to be rotated at all but that means you could rotate it left quote or rotate it right yeah I think

that's that's a very good explanation I don't know if I have much to add Dean you wanted to say something well so but I can't I don't know the answer to this but to take what Daniel just said to take the basic concept of being able to take something that for all intents and purposes can appear one way or another and then turn that into how Chris closed the last lecture with we are we are seeing ourselves as both being entangled and also being separable how did we go literally from a three-dimension Vertex to we are both entangled and we're we're not living in a paradox that's just the way it is we're also removing ourselves or separating ourselves at the same time that's what I would like to know because somewhere between three arbitrary lines and complexity is not a paradox something has to be happening and I'm not exactly sure how that transition plays out anybody want to enlighten me can you repeat your question please well how do we how do we go like from at this point of the reference frame description of saying it can't you you can't copy it all you can do is pass it because we think things could be flipped based on internal internal reference frames internal meanings and internal models the thing we choose for context goes all the way out to the conclusion of and uh of Chris's um conversation with us in in number three where he said bottom line is is that it's not a paradox that as Things become more and more complex then it it wants to be able it it wants to get to a place of entanglement alignment a sense of agreement and at the same time wants to remain separate so between those two things a whole lot has happened but and I don't think we've skipped over it but we certainly haven't opened it up and really looked at it closely and I'm just curious what's in between those two scales I I agree I think we're we're gonna cover that towards the end of the of the talk um but I agree I I don't know how you see this related to qrs uh or the stuff we just talked about but I think it's the big question in this in this lecture right that was like where does this tension between increase in complexity and you know wanting wanting to predict what the environment is doing what one requires separability at least colloquially speaking I I think this tension is basically what the free energy principle comes down to right Daniel what do you think you have the free energy and you have the $E - TS$ one is prediction error I mean not exactly $e - TS$ right that would be thermodynamics but you know one is prediction error and the other one is what was it I can't remember I'll definitely look forward to the the technical unpacking that that you'll get to soon but but I was reminded of the William Blake quotation do what you will this Life's a fiction and is made up of contradiction so let's modify that um project what you will rotate what you will space times a fiction not that it's false but that it's a projected fictionalization that it doesn't have the ontological Primacy that the topology of communication does and is made up of contradiction

yes and it's made up of this tension between fitting and not overfitting between aligning and the principle of unitarity and differentiating and becoming through others we become ourselves and that type of articulation and uh recognizing that two complementary forces like convex and concave come into existence at once like a zero to two jump is not an issue or or a stumbling block it's that we wouldn't have it any other way couldn't have it any other way yeah uh sure we'll get that towards the end uh you can almost even call it a duality it's not uh far-fetched I mean it's it's the sort of thing that I think most people will be happy to take for granted such you know super symmetry like you have two kinds of particles but the same amount of them and you know dualities are all over the place so I I think I think it's fairly it's not unreasonable to just accept that there are there is a tension between uh increasing complexity which you want you want AME to be separable for and accuracy prediction now how this is related to um emergent space and uh on this curves stuff you know I I don't want to make very part fetch claims we'll talk about emergent space later on the course uh but but for now let's let's say a few more words about carefs and then we'll get to the the complexity stuff towards the end so qrf's are almost building blocks if you want or information processing uh here it's very uh useful to recall definition that I feel I think we we have mentioned before of information as being differences that make a difference that's probably the best colloquial definition of information I've heard and I think I think it does a lot of work I mean it's almost mathematical but anyways we need to unpack it a little more and and see so for a difference to make a difference what do you want out of qrf right and surprisingly I mean not surprisingly but reassuringly I think would be the work you need a qrf to have sensory action and a default state right this is sort of like uh where your basing prior distribution is encoded these three things we'll give you a building block for any information processing and that you know ties together with all the work in the free energy literature and perception action loops and and whatnot um so before we see mathematically poetry RF is defined uh I think it's more illustrative and I think Chris went exactly on the right order when when he presented this a couple weeks ago teen uh examples from age these are actual pieces of forward that are effectively curios and it's it's remarkable how they map to well I'll skip ahead a little bit but when you get to the to the mathematical definition of the qrf as a Quantum system that um you know you can specify with with this category Theory diagram you know you have incoming arrows into a node that then they expands right and this graphic correspond to measurement and preparation degrees of not degrees of 300 degrees of freedom would be the same right when you attach it to the boundary but measurement and preparation stages I guess would be

the the better word ah but anyways let me go back to to these examples right um and Daniel your the biologist here so correct me by saying anything wrong here you have uh uh cartoon picture of a bacteria I think this is a very simple sort of thing right it navigates in in the direction of food and and that's it right so it is a very simple rudimentary example of a curve you have the sensory stage here I guess the degrees of freedom I guess the degrees of freedom um it's not obvious here what the boundary is that's one thing uh but it is a good enough picture to see the resemblance to here we have the measurement which in this in this case you're trying to measure concentration of food you have your node here scene which is your prior distribution for concentration of food and finally you have the the preparation stage which is your you know this the rest of this chemical Network that will control uh and steer this bacterium right so there's here you want to call it the action degrees of freedom um that will steer this bacteria in the direction of food and likewise with cell division this I understand even less the network appears more complicated uh but I think it's fair to say that at the end of the day all these chemical Networks that appears super complicated somehow have to be mapped to this category theoretical diagram that's that's all there is to it you're going to have the sensory input and the action output again you're gonna have to the default chemical concentration which amounts to this to this node over here you remember this from last lecture right Daniel yeah in the cone coon we explored also in in live stream 17 on the Bayesian semantic information flow but I'll kind of add a point about these biological systems um in relationship to to artificially specified systems and what that helps us understand about perception cognition action so the cone coone diagram has this bow tie architecture or this kind of like two pyramids touching point to point where there's a preparation and a measurement which can be um different dimensionalities or or different Scopes like you could be multi-input multi-output or multi-input single output vice versa and so on but they come through this eye of the needle with C Prime here now contrast that kind of architecture um represented here abstractly with what we see in the bacterial process and in the cell division process these have that kind of a bow tie perception cognition action architecture or measurement cognitive integration preparation architecture in contrast we see for example an artificial neural network where the input and the output layers are usually lower dimensional maybe uh in an image coming in with several hundred pixels and a single output like a classifier coming out the other end but a dimensional explosion in those internal cognitive layers but in biological systems it's actually exactly the opposite for example many many many receptors on the cell surface on the holographic boundary the sensorium of the cell but

then there are these common secondary Messengers that many Pathways converge upon so that reflects the kind of radically different architecture that a qrf embodies as opposed to the approach taken by disembodied intelligence Frameworks which use dimensional explosion internally to try to categorize and classify and find edge cases versus this dimensional winnowing that is reflective of these embodied intelligences anybody make a tangential comment here also no I think the good thing about these two examples is that they also showcase the fact of how you know the the Deep implications of scale free biology if you want to call it this way we tend to think of these sorts of processes right of such as here only happening at the level of the nervous system in animals or as you said neural nerves right we tend to associate this with neural networks and and perhaps Evolution has tear does in a way that's um endowed us with nervous systems to make these things maybe these are higher up in the in the in the layers uh but really there is you know that's a bit of an arbitrary distinction that we make you know the process is going on at the cellular level even if they don't appear to have anything to do with the nervous system my understanding is that uh they are there are there are some generic properties that are the same and everything that has to do with information processing you know it manifests maybe a slightly different uh when it comes to explicit nerves but at the end of the day it's the same generic process going on would you would you say that's right Daniel the the ontological claim is that it's the same generic process happening whereas the more instrumentalist claim is that we can model it with some shared abstractions yeah okay so we can we can continue from you know I I don't have a whole lot to say Daniel said quite a bit about at the end of the day craft is is you can visualize it as this diagram and you know you can think of it as if you are more familiar with a picture of a neural network and ways on a neural network and so on that's the way to think about it now let's continue with explicit embodiment if you wish uh so qrf's in in the formalism have to be attached to a boundary boundary degrees of terms and I should add another remark here I think and personally to me also like the hard part about uh this Quantum free energy principle formalism if you wish is is what you said at the beginning is the fact that it's topological right so so very often it's not clear what the boundary degrees of freedom of right because we tend to think of boundaries as you know geometrical and certainly the classical fep's the most geometrical uh that's not the case here in the quantum for energy uh formalism and where we have emerging space with us what we talked about in later sessions but that's just like one one quick remark to make at the end of the day though even if it's topological we're going to have real degrees of freedom that um these your ref is attached to this cone cocon diagram and

you're going to have measurement operators that you don't um classical outcomes plus and minus one and then you can think of this PRN um or more abstract like this diagram as as a neural network if you wish acting on those under qubits and then back propagating some sort of action signal preparation um okay then let's not forget that uh Land hours principle and so on requires that we are mindful of energy constraints on computation and that gives us the following picture uh number one we're gonna have to use energy from the environment to fuel our computation to fill our observation so that means that right away you cannot use the entirety of your boundary to gather information you're there's it necessitates a uh an allocation of resources right here e is the observed environment and the blue portion is a thermodynamical sector where you extract free energy and and dump key in order to run this computational process and then furthermore you can have uh a memory sector where you write things on a boundary okay and these takes up energy also right driving information on onto the boundary by Land hours principle will take up information and uh critical is the minimal interest in observer because it has a memory sector and why minimal here uh implies that we have at least minimal interesting implies that we have more than one layer here the process of writing things into the into a memory sector requires us to have a clock so that we can label those memories um in time okay so we do have time as part of this formalism just not space yet okay no time has to be internal to the help server just just one piece on that that Chris mentioned um as and as you echoed time here is an attribute of some cognitive systems in terms of their internal understanding so that they can sort memories and understand what came earlier or later it's not something that's a priori defined externally which is the case in the kind of space time first now how do we fit communication into here we flip that we're talking about the topology of communication as ontologically primal and now certain kinds of sophisticated agents that's not like necessarily true of a rock or a metronome or a particle of a gas but some kinds of cognitive systems are such that they are able to generate an internal clock concept and that doesn't have to be the same thing as what's happening on the wall with Chronos this can reflect a purely internal time Kairos that enables observers or interactors that have different clock structures and then Land hours principle helps us Arbitrage or connect the informational demands to the energetic or the metabolic demands Dean oh so when I when I first listened to the lecture I thought when the when the term or phrase trade-off was raised I I traditionally have thought of tradeoff as kind of a binary if I'm picking this I'm trading off against all the other things that I didn't pick so one or or the other or others plural but when I saw when I saw this and especially when I heard it in the context of things like degrees of

freedom I began to wonder whether it is because it could sit on either side of that boundary it could be a priori or it could be ad hoc or post-talk a trade-off which prior to this conversation I just assumed that a trade-off was one or the other but essentially what this implies is it it isn't even it's still a kind of a degree of of how much weight you put into it and then it you kind of cross a threshold and then you've made your pick you've you've completed your trade-off or did I again did I misinterpret what I was hearing if it's if it's topological as opposed to this the more conventional space-time is this can can a trade-off be both ad hoc and post hoc seemingly at the same time very interesting question let us just let it float and continue on enter all right because maybe because again if it's if it I'll just say this if it if it can be that that would really move us out of the classical Realm that would essentially say that anytime we've made a pick we re we really haven't made a pick there's still a whole bunch of other things that are going to happen that can that can change whatever that action is that we thought we carried out or enacted if we're talking about in in an internet intern internality state which is what I think we want I think we want to be able to change our minds but let me take that one way Dean um the the analytical equations of the action perception Loop don't tell us in implementation like if we're going to write it in code whether we should use action first procedurally to update our observations or whether we enable our observations to to Wag the Dog and then select action at the end of the loop so is it a perception action Loop or is an action perception Loop and those enable a kind of um seesaw where sometimes we want what we learn to change what we do and other times we want what we do to change what we learn and having that kind of a fulcrum or a pivot and then again to connect it to the vertex popping in and out the analytical formalisms inscribed on the screen of the fep like the textbooks they're ambiguous whether we lead with action and have inference follow or whether we lead with inference and and have action follow or or simply without should have inference follow whichever one of the combinations there are um because we s and then the that um trade-off or runoff is broken by specific embodied cognitive agents that that trade-off about the um the procedural Primacy of perception or can't be broken generically right it certainly makes for a lot of waste heat if you're not careful it'd be exhausting a lot anyway okay so um so I I see I I don't know if we can talk about intent um much with this formalism but we can take a very descriptive uh approach to it and um go down here a little more uh actually before I don't want to get up I'm ahead of myself uh what I mean by that is I whether we can put intent or not in this hierarchy um I I don't know I don't know if it matters a whole lot but at least we can talk about it you know operationally and descriptively and and say just like when you have a clock

you know that the clock is just there right like it is it if you don't have the clock you don't have the memory and that's it right uh and likewise here if we don't have attention uh an attention control compartment we just cannot have we cannot play with all these cure apps uh that are non-commutative at the same time if there is no attention control system to begin with no we don't we cannot talk about a system that has non-commutative care apps meaning here one one presumably goes to memory sectors the other one goes to environment side persons and so on uh so so this is certainly a necessity of the fact that we have um different sectors of the boundary being allocated to different purposes such as environmental observation and memory and so on um within the environment it was it's it's worth remarking that here if you if you think deeply about first principles and what it means to observe an entity um well you still come down to these differences that make a difference sort of thing right which involves whenever you're looking at the environment to if you're gonna identify a system you need sort of a background that tells you that puts the system in context right so that you can identify the system as some something distinct I think this again is something that we have talked about in previous uh discussions where the easiest picture here is something like the gauge speed gauge in your car where p is the gauge itself it tells you the speed but it's meaningless without the reference system right outside of outside of the thing that stays invariant right the context the background that Gauge Movie means nothing to you okay so the command come back to this idea of if we think of information as differences that make a difference in terms of first principles these environment sector that we saw earlier has to be further broken down into pointer on reference to States again this is covered uh in depth in the increases papers uh I'm thinking particularly of free energy for generic Quantum systems uh and again uh just to reiterate these attention control stuff uh compartment which is by the way very reminiscent too the hierarchy that was earlier with the clock uh arising fire up in the hierarchy deeper into the layer likewise whenever we have so whenever you're writing something into into a memory you need you need to have that clock whenever you are deploying uncommunicated graphs you're you're gonna need uh attention an attention control compartment and we just described this operationally if you didn't have it then the agent would not be what it is it could not deploy those non-communitive QRS uh so marching on uh we'll just end here with uh pure app alignment on the free energy principle just a few more words this is possibly the most interesting picture in this in this uh presentation and just we can recall here uh possibilities for qrf alignment and how through energy comes into play in this in this um formalism so is that recall we call a the true real interaction we call this a thermal interaction and and then you

have a bunch of possibilities right and depending on each possibility you're going to see things as uh notes or as in the paper says non-local given variables uh the last situation is what we call alignment when we have qrf's of the same size perfectly aligned okay to attack by recall I think this this would be the situation where you have uh noiseless classical uh computation right if you have a and b just interacting and uh it also also this is the situation where this is this is entanglement right asymptotically once all the carers are aligned you you have entitled and now we can type but maybe to Deans earlier so these two these two slides conclude the discussion basically and then we can open it up um so given this possibilities for how qrf's can be aligned here is deployed by a become via line uh and how you know in each situation you're going to get noise or whatever then what does the free energy principle say in this Quantum setting minimizing variational free energy leads you to entanglemental okay so if theorem here if you're in a picture of B here and the qrf's are aligned uh A and B are going to be intact now there is a tension between this tendency towards alignment uh us the pre-energetic principle will carry you and uh observe reality which is that we're not just a blob of entangled particles there is separability systems are attention between maintaining separability um and at the same time being correlated enough to the environment just to be able to predict predict it or come and now we can maybe go into a more philosophical discussion about the free energy principle um I should also say in the last month I just like I talked earlier at traps it led me into a rabbit fall of looking into his recent work and he's talked a lot about complexity uh I'm being very speculative here so I if someone can call me out uh that would be good in the comments but Justice coming through some of his papers I saw a monograph of 2018 where he talks about complexity and you know he puts forward the second law for complexity in Quantum systems and increased complexity and it is very reminiscent of what's going on here you have a tension between uh systems tending to entanglement beyond the other side of the free energy function well if you wish were systems increase in complexity and that necessarily requires uh that A and B are separate So Daniel do you have any comments at this point yeah there's a lot to it I I like that you showed the four possibilities in the image above and and said it was a one of the most interesting figures I I agree and uh how to connect this to the classical fep and also how to see the quantum fep not as just uh a second story built on top of the classical fep which would be kind of like a a historical chronological understanding of the theoretical apparatus but it's almost like the quantum fep was like a basement we didn't know about because it it gives a broader context and though it appeared conceptually several years later in the timeline we understand that doesn't give it a secondary status so what it really

means for the classical fep to be kind of like a special case of this actually more generic setting there's just a lot I'm very uncertain I can I can I I can wrap it up underneath or you want to say something um we can continue with that but let me conclude and this is all I have to say for today I I think I mean I'm sure Chris will explain this more in coming uh sessions and lectures but I think the main key differences is in is in the emergence of of space how it might emerge from something more fundamental maybe a tensor Network or who knows uh but that's that's the big big difference right and and I suppose the quantum free energy principle generalizes the classical for energy principle because he doesn't have this special embedding now uh this to me personally and and also to folks in the audience for asking questions is is probably the hardest part to wrap your head around and I I foresee interest in in those uh discussions and lectures that deal with the concept of emerging space because every time you try to connect on it I've done this myself I've been thinking about how you could apply these formalism to better known pictures or compare it for instance with thermal equilibrium with relaxation with thermal equilibrium and then you start thinking about okay what is the boundary and so on it's far from obvious right because again you have to remind yourself that the definition is topological so in that regard I think it might be worth one of the questions that came up that I thought I thought was good and and then I can call it a day on this end uh so this question and I think it was the first question asked since the last lecture can you just read the full question and then read the full answer thank you the question on the answer it's number 28 here can you guys see it should I zoom in a little bit it's it's pretty small but but reading it will be good okay so this question I think I've asked myself this question two um maybe with different in a different way but basically you know it gets to the point of what I was saying earlier um that it would be nice to see how we can use this Quantum free energy principle formalism to recover maybe from your settings such as relaxation to thermal equilibrium via ticket equation but that's far from movies how you do it if your definition is topological right so the question says what happens when the agent becomes identical to the environment when the model has been improved so much that it does not differ from the environment is this a stadium impossible would it be thermionamical equilibrium maximum entanglement okay so clearly it's not thermodynamical equilibrium but I think what the person is asking if I maybe if I understand this correctly they're wondering okay so it's a free energy drives AME towards entanglement then it's referred to say that they have that any distinction between them has to be solved right that they have merged into the same thing and that has to do with this whole definition of philosophical evade of thinness that uh called Princeton you

know it's a proponent enough right what it means for two things to be separate I'm coming so let's see what Chris said Kristen has pointed out that the limit of the classical fep the limit of perfect prediction corresponds to generalized synchronic between system and environment each sends messages but the other can perfectly predict uh now however it's clear and I'm not reading anymore I'm making a quick remark here I'll continue in a second in the classical fep it's clear that A and B even if they are you know in the perfect limit of the fep they remain different the different entities right because you have that if you wish that extra uh layers of differentiation coming from their spatial embedding right in the in the classical fpp that is clear that they haven't even if they are getting synchronized they have not lost their separate identities you can always say well this is the one that was to the left and this is the one that was to the right pun intended with earlier discussion right you have that extra uh resource for for separating them if you wish by virtue of the special embedding right the special embedding is giving you this extra information that allows you to keep track of of of amb separately okay so I'll continue this notion depends on the system and its environment being separated and has distinguished somehow In classical formulations they are separated in ordinary 3D space different locations so they are different because they occupy different locations right but if you don't have that geometric wall background then then how you know that they're different right okay so I continue the quantum formulation is background free meaning it assumes no space-time embedding system and environment do not have different locations even though they have different sets of degrees of freedom in the overall Hilbert space here the fpp the limit of the fep is maximum and time moment this does not mean that the system and its environment are identical they aren't since they have different degrees of freedom okay so this is just addresses the question I I just post it means that the joint state is not separable um into a system s an environment team their states are no longer conditionally independent this is distinct from thermal equilibrium which means that they have the same temperature while remaining separable uh and that their interaction can be characterized as the change of thermal fluctuations I.E knows so I guess this is a nice format for things that are to come but I I do think that this or idea of emerging space is going to be possibly you know one of one of the most important intercourse stick out between the classes and the quantum formulations for sure Dean what's your take on this what what I saw what I saw was that when you have that kind of alignment that perfect alignment that is one that was one that was D in that diet in that that diagram right that was one state of a of a complex state that included an A and A B and a c and probably a whole bunch of other ways that yeah right so that one state where you've got

this kind of entanglement isn't isn't into perpetuity like you don't get there and then stay there and that's where I think the question of separability and complexity it still works because just because you momentarily got into a place that we Define here as D doesn't mean you stay there indefinitely you could be swapping around to B and C and A in the next instant so that's that at least that holds the complexity piece up that that moment for however long that moment is where XA and x b are aligned isn't permanent so it's it's not predictable in that sense therefore it is complex I still don't know how we get though that the closer we get to agreement the more complex Things Are unless we're just talking about D in this array of a b c and d that goes back to that Quantum reference frame I think like I said again maybe I'm interpreting this wrong but when I message pass the reason why I have to like literally give it over to the other person and let them have it I can't give have them copy what I'm doing is because there has to be I guess that reinforces the separability piece of this as well the fact that I can't give them a copy that I can only give them the information and then they've got to sort of unpack it in their own in their own way of taking it off the the boundary but again I struggle with this I'm guilty as charged but I am I am what I'm trying to do is trying to come up with my own narrative so that what Chris is saying makes sense to me as opposed to make sense to the world which is what Chris is trying to do I'll try with another speculative fiction um there's a screen between let's just say the stage and the audience or whatever it's a holographic screen and on each side of the screen there's some embodied object so that means that it must have um Vault it's a volume and closing object because points and lines and planes are only abstractions every real physical object encloses volume and now a light is going to be coming from behind each object facing the screen and that's this kind of uh cone that we're seeing from each side it's like the shadow cast by the object so if the object was a sphere then it would be projecting down to a circle onto the screen if the shape were a cube it could be projecting down and so where the light is positioned is the um perspective that that cognitive entity is taking on its qrf those lights could move so now we have the situation where the two spotlights are moving maybe the shapes are staying the same maybe they're changing too and it's like two searchlights scanning across this B oval and some sometimes they are perfectly locked I mean there could be a situation where they're perfectly aligned or there could be situations where they're totally disjoint um and each of those situations have different implications for what one entity sees when it looks at the screen maybe like the shadow is the only part where you can see because the rest of it is too blinding or something like that and so then if we're in case a here where um maybe B doesn't have a projection or maybe B's

projections totally somewhere else then a is in this solipsistic I'm the active agent interacting within an art World setting and D is like potentially the um the computer Engineers classical dream or asymptote which is like all the data going through the USB cable is just being perfectly noiselessly received and then we also get these um B and C cases but a is kind of like the inner world D is like the perfect Communication channel and then to connect it just to friston's concept of generalized synchrony as we've talked about many times generalized synchrony doesn't mean lockstep two agents could reduce their uncertainty about each other not by doing the same thing at every time step but for example by turn taking and I think that points the way to how we get reduction of variational free energy reduction of prediction error surprise and so on without especially spatial temporal enmeshment but not necessarily even informational identity in measurements because you could have two things that through their alignment become actually differentiated and that could increase the mutual information between two metronomes not just to have them going in the same direction but to have them going in opposite directions the mutual information in those two settings is going to be high so it's not the case a generalized synchrony means that um things are becoming blurred we could actually also have integrated information through increased articulation not just decreased and then life finds a way or cognition finds a way on this trade-off Frontier of in our pursuit of unitarity will we blur to unify or will we differentiate to unify but those are the paths we have we don't have the um backwards ratchet yeah makes sense any other ladder slides enter uh no that's it uh so Chris made a few more remarks at the end about uh language here uh but you mentioned that he's gonna talk more about this in the coming lecture so we'll just leave it up to him yeah there could be some very powerful and interesting linguistic opportunities for updating our speaking to reflect what we are learning here I don't know what those necessarily are whether it's a syntactic or a semantic thing or whether it's a narrative thing but how to actually re-enter some of these formalisms into our Focus understanding could just be immensely powerful and um we could have a lot of Integrity between the natural language way that potentially we're already showing up for engagement and some of these formalisms gets a first principles about interacting entities and boundary conditions and all of that Dean any penultimate thoughts not really I I'm going to go back and listen to the I'm going to listen to Chris's comments again about this common cause and the fine-tuning Assumption and whether that's sort of a metaphorically a dialing in or what it what he means exactly by the fine tuning in the in the context of this conversation today what you just kind of said about are we dulling something here or are we sculpting something to make it more

fittable like I'm going to go and now and go back and listen to what he talks about in terms of fine tuning and maybe update my priors around that so enter any last uh thoughts or maybe scroll over to the um schedule the course again just so we can close on yeah we're halfway through we just completed the third discussion section now so we've had three lectures and three discussions it's all downhill from um by the way this was the first midterm you guys both did well um we're gonna head into four or like session four with these last three how agents use multiple communication channels and that's going to get connected to the quantum fep how space time emerges from communication which is something that we've brought up a bunch it's been in the background but but making That explicit it's going to be exciting and then section six I think is just gonna be such a jumping off point because the scope and the intensity of what is transpiring is Extreme so connecting it to biology which is to say life more fully and where we go in in 23 and Beyond that's going to be awesome so any last words Ender uh no I just think of what you said um very I would I would say it's going to be uphill in in terms of how interesting it's going to get cool well I encourage people to uh check out the course website and submit questions that they'd like to see addressed by Chris also to register and join these participatory discussions there'll be three more of them so if you want to join share your qrf be part of the publication of all of these works that will compile at the end and just get your exoskeleton in the game with what is happening here it'd be really awesome to hear more voices and and to have more on board okay thank you Dean and Andrew till next time